

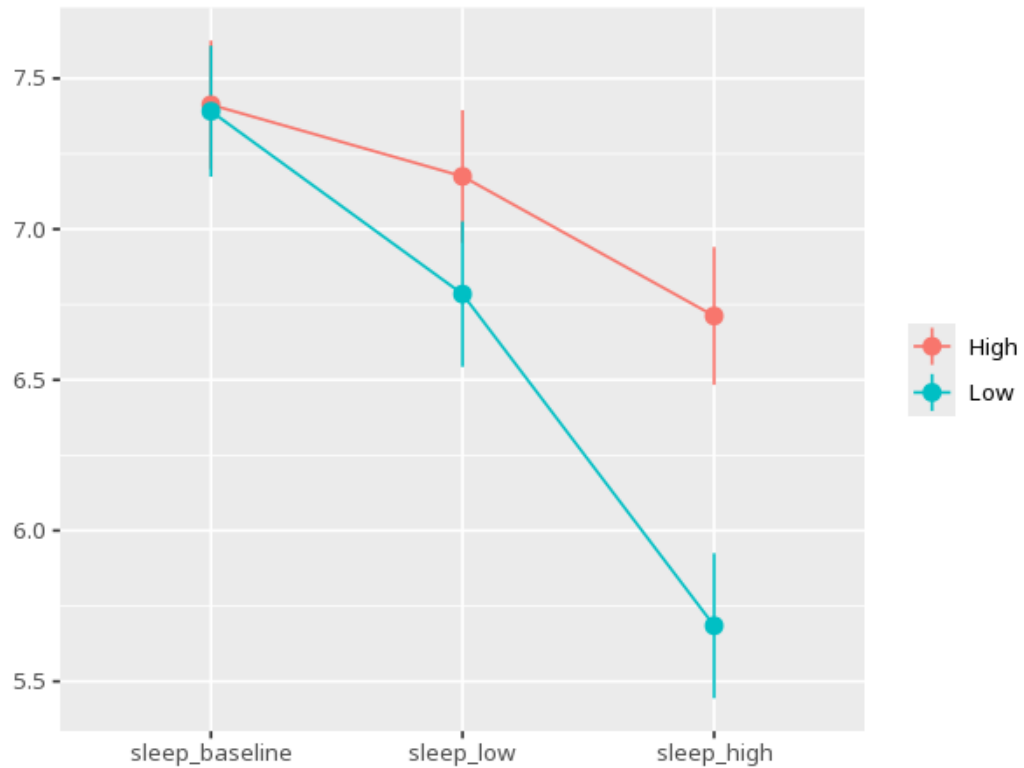
1 Mixed ANOVA

Group	Condition	N	M	SD
High	1	20	7.41	0.46
Low	1	20	7.39	0.46
High	2	20	7.17	0.47
Low	2	20	6.78	0.51
High	3	20	6.71	0.49
Low	3	20	5.68	0.51

Source	SS	df	MS	F	p	partial η^2 [95% CI]
Tolerance_group (between)	6.91	1	6.91	10.37	.003	0.21 [0.05, 1.00]
Condition (within)	29.83	1.60	18.60	759.23	<.001	0.95 [0.94, 1.00]
Tolerance_group $\times$ Condition	5.18	1.60	3.23	131.84	<.001	0.78 [0.70, 1.00]

Effect	W	p	GG ε	HF ε
condition	0.75	.005	0.80	0.83
grp:condition	0.75	.005	0.80	0.83

Pair	Mdiff	SE	padj	95% CI
sleep_baseline - sleep_low	0.42	0.03	<.001	[0.36, 0.49]
sleep_baseline - sleep_high	1.20	0.03	<.001	[1.13, 1.27]
sleep_low - sleep_high	0.78	0.04	<.001	[0.69, 0.88]



Read the Group $\times$ Condition interaction first - a significant interaction means the groups changed differently across conditions; only then read the main effects. Check sphericity for the within and interaction terms - if violated, read the corrected F/p. When the group $\times$ condition interaction is significant, the overall condition comparisons in the post-hoc table can mislead - read each group's line on the profile plot instead. Your significance threshold (α) is 0.05.

A mixed ANOVA yielded a Tolerance_group $\times$ Condition interaction, $F(1.60, 60.96) = 131.84$, $p < .001$, partial $\eta^2 = .78$ [.70, 1.00].

Statistical basis: Mixed (split-plot) ANOVA (Fisher, R. A. (1925). *Statistical Methods for Research Workers*. Oliver & Boyd.); Mauchly's test of sphericity (rendered above the correction) (Mauchly, J. W. (1940). "Significance test for sphericity of a normal n-variate distribution." *The Annals of Mathematical Statistics*, 11(2), 204-209.); Greenhouse-Geisser correction (sphericity violated) (Greenhouse, S. W., Geisser, S. (1959). "On methods in the analysis of profile data." *Psychometrika*, 24(2), 95-112.); Huynh-Feldt correction (sphericity violated) (Huynh, H., Feldt, L. S. (1976). "Estimation of the Box correction for degrees of freedom from sample data in randomized block and split-plot designs." *Journal of Educational Statistics*, 1(1), 69-82.); Levene's test for equal variances between groups (Brown-Forsythe median-centered variant) (Levene, H. (1960). "Robust tests for equality of variances." In I. Olkin (Ed.), *Contributions to Probability and Statistics* (pp. 278-292). Stanford University Press.); Brown-

Forsythe median-centered variant of the homogeneity-of-variance test (Brown, M. B., Forsythe, A. B. (1974). "Robust tests for the equality of variances." Journal of the American Statistical Association, 69(346), 364-367.); Estimated marginal means for post-hoc comparisons (Lenth RV (2024). emmeans: Estimated Marginal Means, aka Least-Squares Means. R package.).